

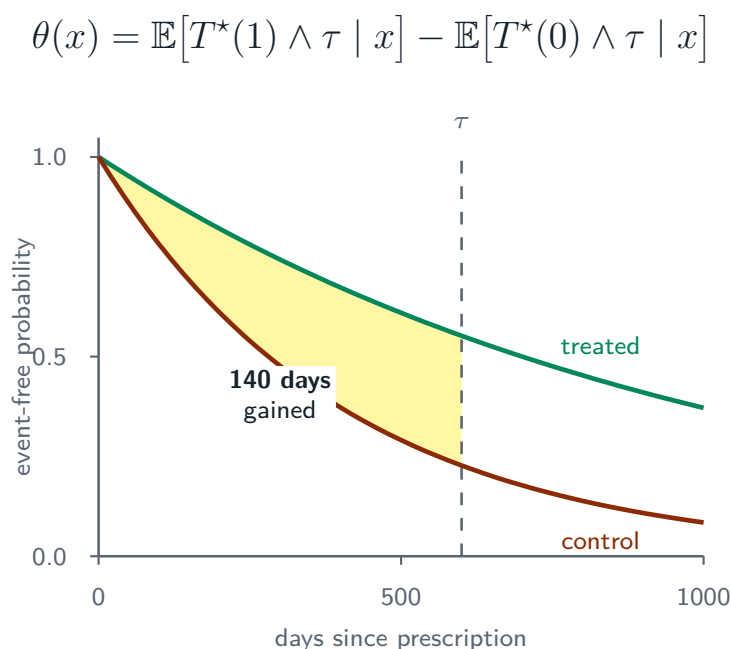
What Censoring Costs a Doubly Robust CATE Learner

An exact finite-sample error budget for treatment effects on survival outcomes

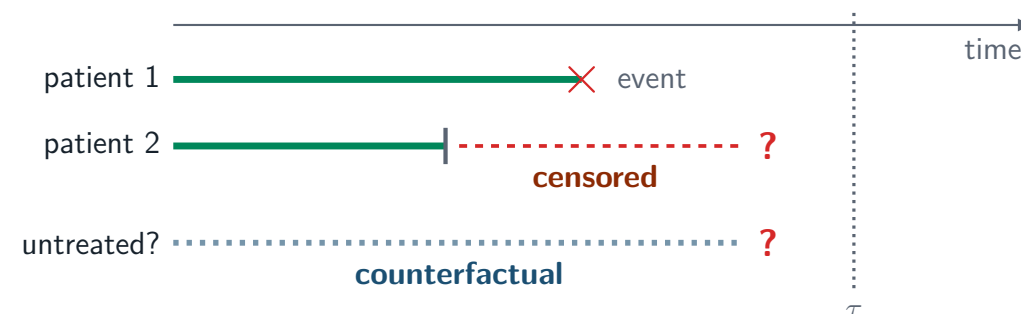
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Estimand: RMST gain



Two missing pieces

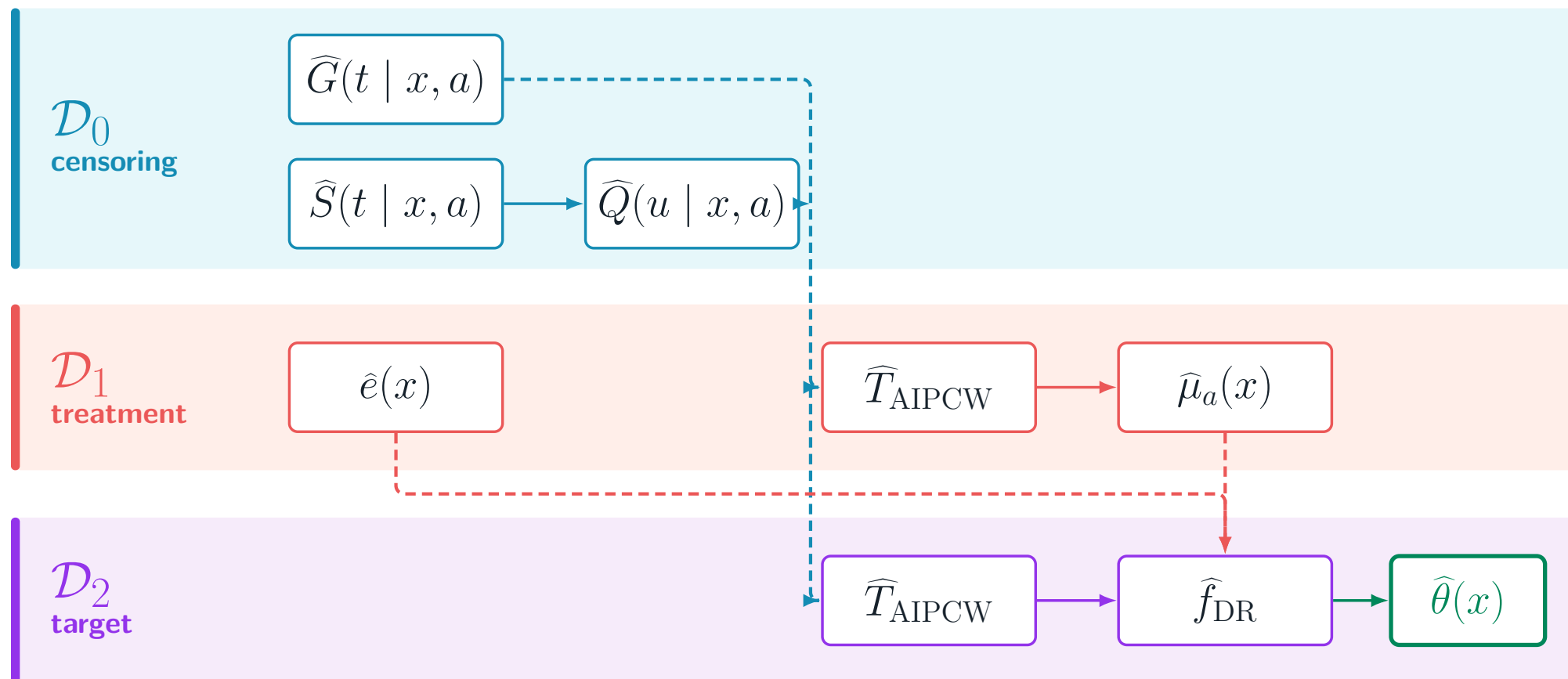


confounding
DR pseudo-outcome
 $\eta_T = (e, \mu_0, \mu_1)$

censoring
CUT (AIPCW)
 $\eta_C = (G, Q)$

$$\eta_T \cap \eta_C = \emptyset$$

Three-fold AIPCW-DR



🎯 The bias, exactly

$$\begin{array}{c} \boxed{\hat{b}(x)} \\ \uparrow \\ \text{TOTAL BIAS} \end{array} = \begin{array}{c} \boxed{\mathcal{R}^{\mathcal{D}_0}(x)} \\ \uparrow \\ \text{CENSORING REMAINDER} \end{array} + \begin{array}{c} \boxed{\frac{(\hat{e} - e)(\hat{\mu}_1 - \mu_1)}{\hat{e}} + \frac{(\hat{e} - e)(\hat{\mu}_0 - \mu_0)}{1 - \hat{e}}} \\ \uparrow \\ \text{ORTHOGONAL BIAS} \end{array}$$

AIPCW: the remainder is a product

$$\delta(x, a) = \int_0^\tau \frac{S G}{\widehat{G}} \underbrace{(Q - \widehat{Q})}_{\text{imputation error}} \underbrace{(\mathrm{d}\widehat{\Lambda}_C - \mathrm{d}\Lambda_C)_{\text{censoring error}}}$$

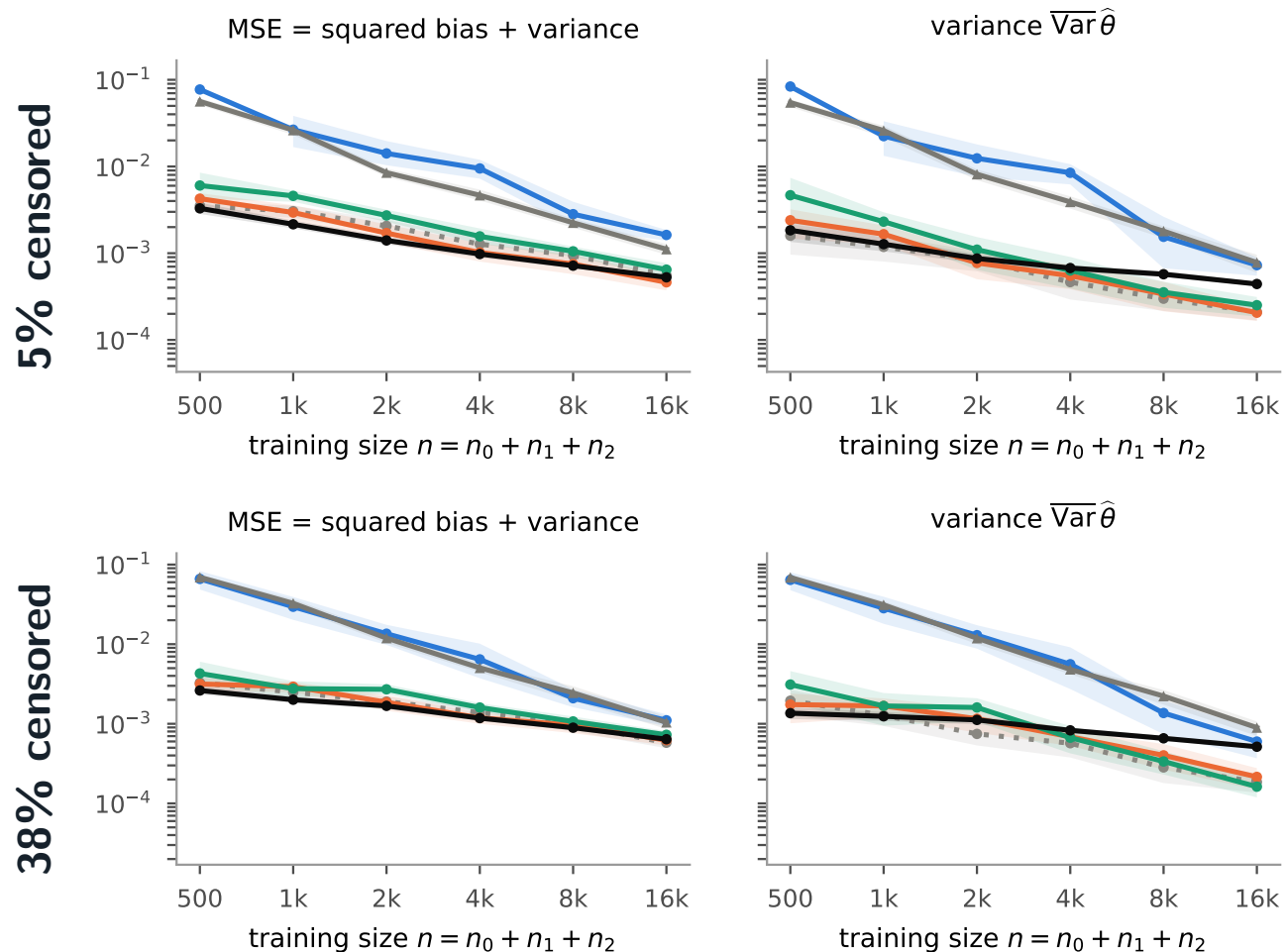
The variance, exactly

$$\begin{array}{c} \boxed{\hat{v}(x)} \\ \uparrow \\ \text{TOTAL VARIANCE} \end{array} = \begin{array}{c} \boxed{\frac{e}{\bar{e}^2} \bar{\sigma}_1^2 + \frac{1-e}{(1-\bar{e})^2} \bar{\sigma}_0^2} \\ \uparrow \\ \text{DISPERSION} \end{array} + \begin{array}{c} \boxed{e(1-e)(g_1 - g_0)^2} \\ \uparrow \\ \text{ARM IMBALANCE} \end{array}$$

AIPCW: $\widehat{Q}$ is a variance nuisance

$$\hat{\sigma}_a^2 = \sigma_a^2 + \int_0^\tau \frac{S}{G} \left[\underbrace{v}_{\text{residual-life variance}} + \underbrace{(\widehat{Q} - Q)^2}_{\text{imputation error}} \right] d\Lambda_C$$

Benchmark on finite-size datasets



..... AIPCW-DR, oracle nuisances —●— AIPCW-DR, forests (grf) —●— CSF (Cui et al., 2023)
 —●— AIPCW-DR, linear —●— AIPCW-DR, boosting —▲— SurvITE (Curth et al., 2021)